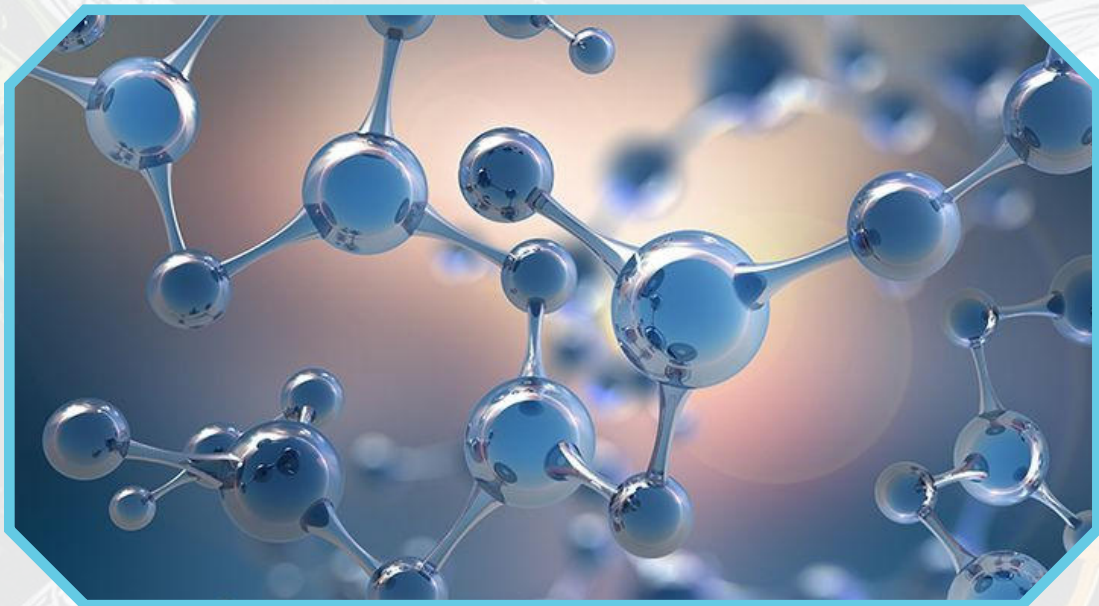




QUANTUM PHYSICS



Introduction:

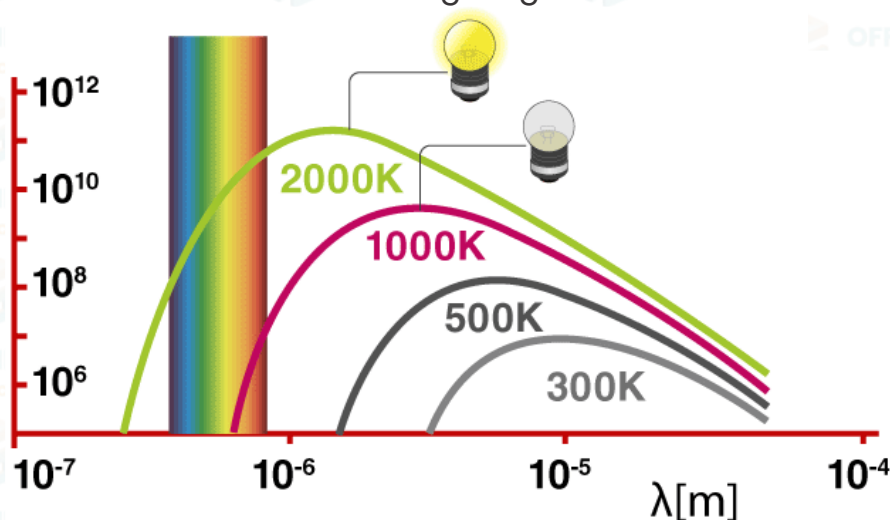
Quantum physics is the study of matter and energy at atomic and subatomic levels. It aims to discover the properties and behaviors of the very building blocks of nature.

Quantum Theory of Radiation:

It was introduced by Max Planck and Albert Einstein. It explains how energy is emitted and absorbed in discrete packets (quanta), the key points of Quantum theory of radiation are:

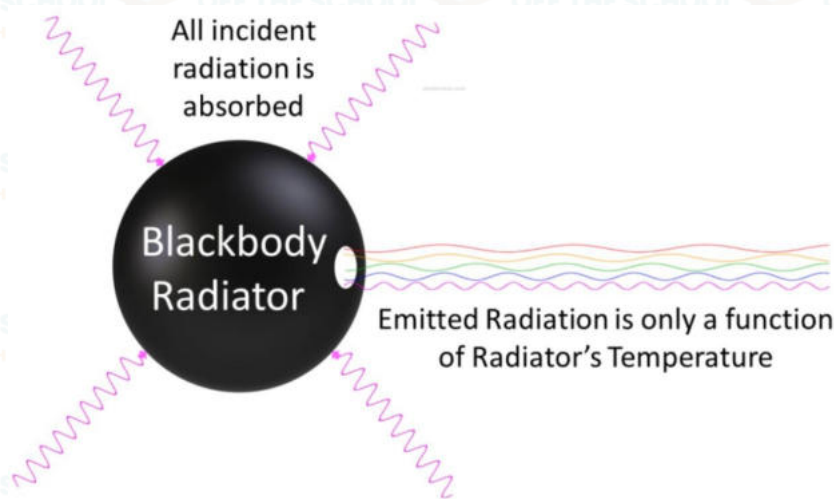
- Energy is quantized (small packets, not continuous).
- Frequency (not amplitude) determines energy of radiation.
- Radiation has both wave and particle properties (duality).

This theory revolutionized our understanding of light and radiation.



Black Body Radiation:

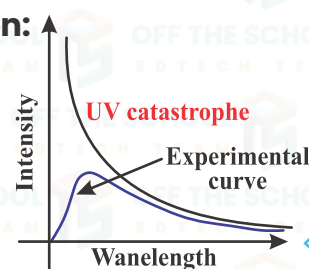
A black body is an idealized object which absorbs all incident radiation and also emits it in a continuous spectrum of colors, depending upon its temperature.



Classical explanation of black body significant challenge to radiation:

Rayleigh-Jeans Law:

This law states that the energy per unit volume per unit wavelength of blackbody radiation is inversely proportional to the fourth power of the wavelength (λ), not directly proportional to the square of the wavelength.



The law is given by the formula:

$$E(\lambda, T) = \frac{2cK T}{\lambda^4}$$

Mathematically expressed as:

$$E(\lambda, T) = \text{Constant} \frac{1}{\lambda^4}$$

where λ is the wavelength

Wien's Displacement Law:

This law breaks down completely at low frequencies.

Wien's displacement law states that the wavelength at which the intensity of blackbody radiation is maximum is inversely proportional to the temperature of the blackbody.

As the temperature increases, the peak of the blackbody radiation curve shifts to shorter wavelengths. This relationship is mathematically expressed as:

$$\lambda_{\max} T = \text{Constant}$$

Where $\lambda_{\max}$ is the wavelength at which a blackbody radiates at a given temperature T .

Stefan's Law:

The experimental relation is known as Stefan Boltzmann's law;

It states that the total energy radiated per unit surface area of a black body is proportional to the fourth power of its temperature.

Mathematically, it is expressed as:

$$P = \sigma A T^4$$

Where A is the surface area of a blackbody, T is its temperature, and σ is the Stefan-Boltzmann constant,

$$\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$$

Planck's Hypothesis:

Planck proposed that electromagnetic energy could only be emitted or absorbed in discrete packets, which he called "quanta." This assumption led to the concept of energy levels being quantized, meaning that the energy could only take on certain discrete values. The energy (E_n) of an oscillator in the cavity walls is given by the equation:

$$E_n = nhf$$

Where:

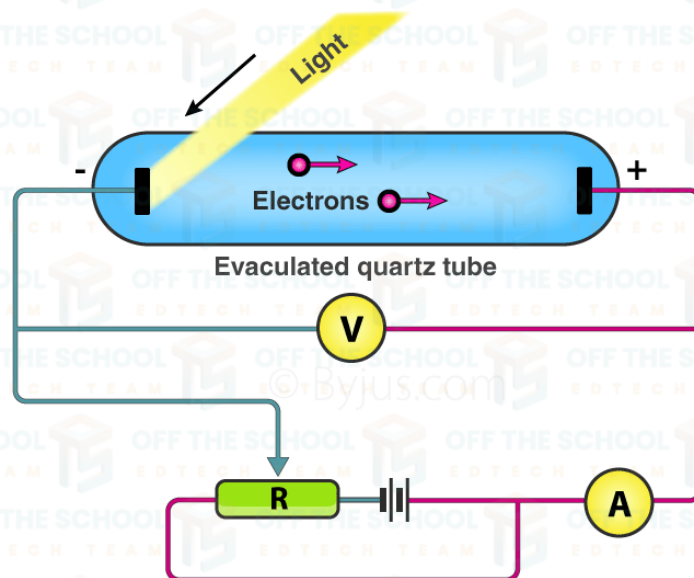
- E_n is the energy of the oscillator,
- n is an integer (1, 2, 3, ...),
- h is Planck's constant,
- f is the frequency of the oscillator.

Photoelectric Effect:

When electromagnetic radiation like light is shined on certain metallic materials, electrons are emitted due to absorption of light by the electrons on the surface of material.

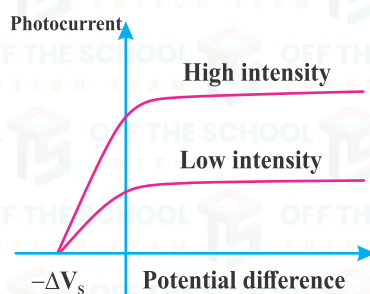
Photoelectric Effect Phenomenon:

The photoelectric phenomenon is shown in figure, in which an evacuated tube containing two electrodes cathode and anode; connected with a variable source voltage V . A monochromatic light source is shined onto cathode through a quartz window and anode (collecting electrode) is connected either positive or negative with respect to the cathode. An ammeter is connected with circuit to record the current due to photo-electrons.



Following observations were made during this experiment:

1. For a constant potential difference between the cathode and anode, the number of electrons emitted from cathode increases with increasing intensity of radiation.
2. For a constant intensity and frequency of incident radiation the photoelectric current varies with the potential difference V between the cathode and anode and reaches a constant value beyond which further increase of potential difference does not affect the photoelectric current; instead, if the anode is made more and more negative with respect to the photo-cathode surface the current decreases. This negative potential (with respect to cathode) of the plate is called retarding potential.



3. For a particular value of retarding potential, the photoelectric current becomes zero. This potential is called cut-off or stopping potential V_0 and is measure of maximum kinetic energy of photoelectrons and we can write:

$$K_{\max} = E_{v0}$$

4. The stopping potential and hence the maximum kinetic energy $K_{\max}$ of photo-electrons is independent of the intensity of incident radiation and depends only on the frequency ν of radiation.

5. For each substance there exists a characteristic frequency ν_0 such that for radiation with frequency below the photo-electrons are not ejected from the surface. This frequency is called the threshold frequency and the corresponding wavelength is called Threshold wavelength,

$$\lambda_0 = \frac{c}{\nu_0}$$

6. The time lag between the incidence of radiation and the emission of a photoelectron is very small, less than 10^{-9} seconds.

Explanation of photo-electric effect on Quantum Theory:

Einstein proposed that radiation energy is not continuously distributed over the wave-front, but the light energy consists of discrete quantum of energy $h\nu$, which penetrates the surface of the cathode, all of its energy is transformed to an electron; it's depicted. The photon's energy would be associated with its frequency ν , through a proportionality constant known as Planck's constant h , alternatively, using the wavelength λ and the speed of light c :

$$K.E = h\nu = \frac{hc}{\lambda}$$

or the momentum equation:

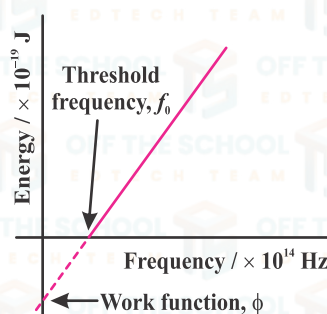
$$p = \frac{h}{\lambda}$$

When a photon is incident on the surface of a material, some of its energy is spent in making the electron move and the rest appears as kinetic energy of the electron. The electrons at the surface for the material are most loosely bound and require minimum energy for the exit liberation.

This energy is called the work function ϕ of the material. The maximum kinetic energy of photo electrons ejected from the surface is given by:

$$K.E_{\max} = \frac{1}{2}mv^2 = h\nu - \phi$$

So, work function is the minimum energy that must be supplied to the electron for it to leave the metal surface.



Consequently, there exists a minimum frequency that is independent of the intensity of light, below which electrons cannot be ejected from the metal.

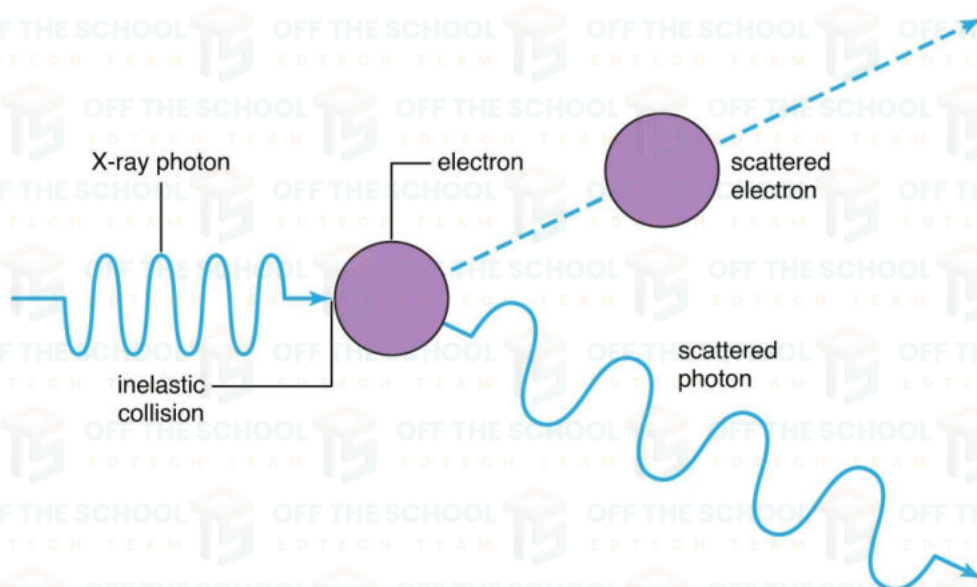
This proposal was proved by Millikan when in 1914 he published his results of the voltage required to stop photoelectrons ejecting from a metal surface.

Photoelectric effect in solar cells and photocells:

- Photovoltaic cells are made up of p-type and n-type silicon semiconductors. When light falls on the junction between p-type and n-type semiconductors, electrons are emitted according to the photoelectric effect. These electrons are collected in an external circuit to produce an electric current.
- o **Photocell**, a device that uses the photoelectric effect to generate or control a current.
- o **Photovoltaic cells** and **photoconductive cells** are examples of photoelectric cells.
- o **Solar cell**, a device that converts electromagnetic energy into electrical energy.
- o **Photovoltaic cells** are used to power a wide variety of devices, and are often used in conjunction with a battery that stores the power they produce.

Compton Effect:

He observed that the scattered light had a slightly longer wavelength than did the incident light, and therefore a slightly lower frequency indicating a loss of energy. He explained this result based on the photon theory as incident photons colliding with electrons.



Compton Effect Qualitatively:

Compton's Effect, as illustrated, can be qualitatively explained as follows:

Initial Interaction:

A photon, which is a packet of electromagnetic energy, interacts with an electron in a material, typically a target like a metal or graphite.

Scattering Process:

During the interaction, the photon transfers some of its energy and momentum to the electron. This transfer causes the photon to change its direction and wavelength (or equivalently, its frequency).

Change in Photon Energy:

The scattered photon emerges with less energy (longer wavelength) than the initial incident photon. The amount of energy lost by the photon is directly related to the energy and momentum gained by the electron.

Quantum Nature:

Compton's Effect cannot be explained using classical wave theory alone.

Classical wave theory predicts that light should scatter uniformly without a change in wavelength.

However, Compton's observations demonstrated that the scattered light has a shifted wavelength, indicating a particle-like interaction.

Experimental Confirmation:

Compton conducted experiments where X-rays were targeted at graphite and the scattered X-rays were observed. By measuring the scattering angle and change in wavelength, he confirmed that the results were consistent with the predictions of quantum theory.

Wavelength Shift:

The wavelength shift observed in Compton scattering is directly proportional to the Compton wavelength, which depends on the mass of the electron and the Planck constant. This relationship provides crucial evidence for the particle nature of photons.

The Compton shift can be evolved as under:

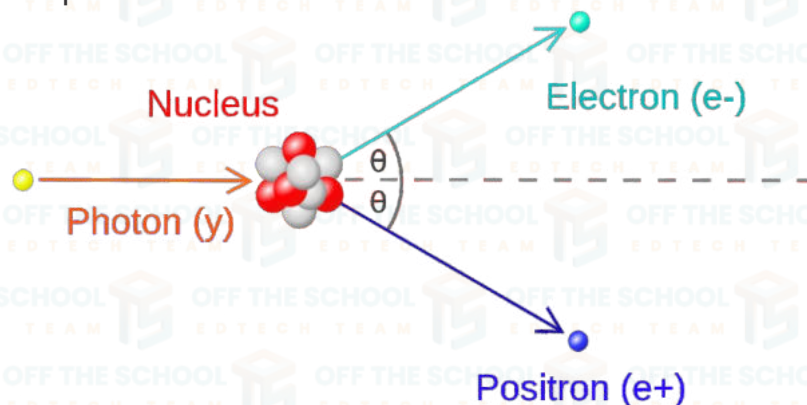
$$\lambda_2 - \lambda_1 = \frac{h}{m_e c} (1 - \cos \theta) = \frac{h}{m_e c} (1 - \cos \theta)$$

$$\Delta\lambda = \frac{h}{m_e c} (1 - \cos \theta)$$

This equation describes the phenomenon known as Compton Effect.

Pair Production:

Pair production is a phenomenon in quantum mechanics that occurs when a photon with sufficient energy interacts with matter and spontaneously transforms into a particle-antiparticle pair.



Conservation laws of Pair-Production:

Facts of pair production process are:

1. The pair production process obeys the law of conservation of energy, momentum and electric charge respectively.
2. During collision, the antiparticle of an electron i.e., positron has the same physical properties as electron, except its charge, as both have opposite charge to each other. The sum of charges happens to be zero which is equal to photon before interaction. Therefore, electric charge is conserved.

3. The law of conservation of energy is antiparticle counterpart.

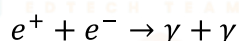
$$(a) hf = (K.E_e) + (K.E_{e^+})$$

$$(b) hf = 2m_e c^2 + (K.E_e) + (K.E_{e^+})$$

Annihilation of Matter:

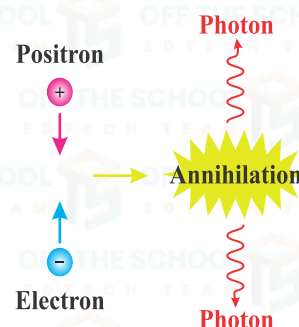
Electron-positron annihilation occurs when an electron and a positron collide. The result of the collision is the conversion of the electron and positron to the creation of gamma-ray photons.

Conservation laws of annihilation of matter:



Each photon has an energy equal to the rest mass of electrons 0.51 MeV.

Pair production and annihilation of matter supports that energy and mass are interconvertible i.e. $E = mc^2$.



Wave Nature of Particles:

The wave nature of particles implies that particles can behave like waves, displaying properties such as reflection, refraction, interference, diffraction, and other wave-like behaviors.

Characteristics. In essence, this duality suggests that matter and radiation can co-exist both particle and wave attributes.

Examples of particles that exhibit this dual nature include:

1. Matter particles, such as electrons, protons, and neutrons.
2. X-rays.
3. Photons.
4. Electromagnetic radiation.

De Broglie wave:

De Broglie's idea suggested that both matter and energy could exhibit both particle and wave properties. This unified description was a departure from the classical distinction between particles and waves, providing a more comprehensive understanding of the behavior of particles at the quantum level.

De Broglie Wavelength:

According to de Broglie hypothesis, all matter particles like electrons, protons and neutrons in motion are associated with waves. These waves are called de Broglie waves or matter waves. The momentum of photon of frequency ν is given by:

$$p = \frac{h\nu}{c} = \frac{h}{\lambda}$$

$$\lambda = \frac{h}{p}$$

According to de Broglie, the above equation is completely applicable to matter

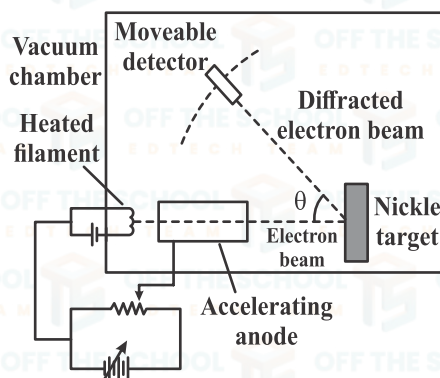
particles as well. Therefore, for a particle of mass m traveling with speed v , the wavelength is given by:

$$\lambda = \frac{h}{mv}$$

This wavelength of the matter waves is known as de Broglie wavelength.

Davisson – Germer Experiment:

They demonstrated that electron beams are diffracted when they fall on crystalline solids. Since crystal can act as a three-dimensional diffraction grating for matter waves, the electron waves incident on crystals are diffracted off in certain specific directions. This shows a schematic representation of the apparatus for the experiment.



The experimental parts are given as under:

1. An electron gun emits electrons via thermionic emission produced by the tungsten filament used in it, i.e., when heated to a specific temperature.
2. Two opposite charged plates known as Electrostatic Particle Accelerator, which accelerates the electrons at a certain potential.
3. The accelerator is enclosed within a cylinder called Collimator which has a narrow passage for the electrons along its axis.
4. The target is a Nickel crystal on which the electron beam is fired normally.
5. When the electrons are scattered from Ni crystal, these are captured by the semicircular movable detector.

Davisson Germer Experiment and de Broglie Relation:

Let us consider an electron of mass (m_0), charge (e) accelerated from rest through a potential V . Then, the kinetic energy K of the electron equals to the work done (eV) on it by the electric field:

$$K.E = eV$$

$$\text{Now, } K.E = \frac{1}{2}mv^2 = \frac{p^2}{2m_0}$$

$$p = \sqrt{2m_0 K.E} = \sqrt{2m_0 eV}$$

Then, the de Broglie wavelength of the electron is given by:

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m_0 eV}}$$

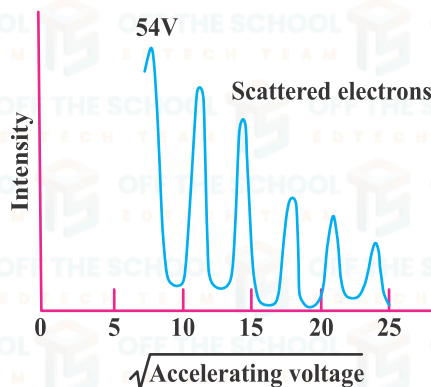
By using:

- $h = 6.626 \times 10^{-34} \text{ J-s}$
- $m_0 = 9.11 \times 10^{-31} \text{ kg}$
- $e = 1.60 \times 10^{-19} \text{ C}$

We get:

$$\lambda = \frac{1.227}{\sqrt{V}} \text{ nm}$$

Where V is the magnitude of accelerating potential (wavelength of electron in volts).



Electron Microscope:

De Broglie hypothesis is used in the field of electron optics. In the electron microscope design, the wave properties of electrons have been utilized with higher resolution, which is giving a great improvement in visualization.

A special type of microscope that uses an electron beam with its wave-like properties to illuminate a specimen able to magnify objects in high resolution (nanometers), which are formed by controlled use of electrons in a vacuum captured on a phosphorescent screen. Ernst Ruska (1906–1988), a German engineer and academic professor, built the first Electron Microscope (EM) in 1931, and the same principles behind his prototype still govern modern Ems.

De Broglie hypothesis of electron have wave paved the way for the development of the electron microscope (EM), which can produce images of much greater magnification than an optical microscope. There are two types of electron microscopes which are transmission electron microscope (TEM), which produces a two-dimensional image, and the scanning electron microscope (SEM), which produces images with a three-dimensional.

Working:

In both types i.e., **TEM** & **SEM**, the objective and eyepiece lenses are actually magnetic fields that exert force on the electrons to bring them to a focus. The fields are produced by carefully designed current-carrying coils of wire.

EM's measure the intensity of electrons, producing mono chromatic image. Color is often added artificially to highlight. The probe tip of scanning electron microscope, as it is moved horizontally, automatically moves up and down to maintain a constant tunneling current, and this motion is translated into an image of the surface.

Applications:

- That of visible light photons, Electron microscopes are used to investigate the ultra-structure of a wide range of biological and inorganic specimens including microorganisms, cells, large molecules, biopsy samples, metals and crystals. Industrially, electron microscopes are often used for quality control and failure analysis.
- Modern electron microscopes produce electron micrographs using specialized digital cameras and fine grain film to capture the images.
- The advancement of microbiology is significantly indebted to the electron microscope, which has revolutionized our understanding of microorganisms such as bacteria, viruses, and their pathogens, thereby greatly enhancing the effectiveness of disease treatments.

Role of Electron Microscope:

1. Higher Resolution: Electron microscope offers significantly higher resolution compared to optical microscopes, allowing observation at the nanometer scale due to the shorter wavelength of electrons.

2. Transmission Electron Microscope (TEM): TEMs study internal structures of thin specimens, producing detailed images of cells, organelles, and crystalline structures. Valuable in biology, materials science, and nanotechnology.

3. Scanning Electron Microscope (SEM): SEMs provide 3D surface images by scanning specimens and detecting emitted secondary electrons. Widely used in biology, geology, and materials science for surface analysis.

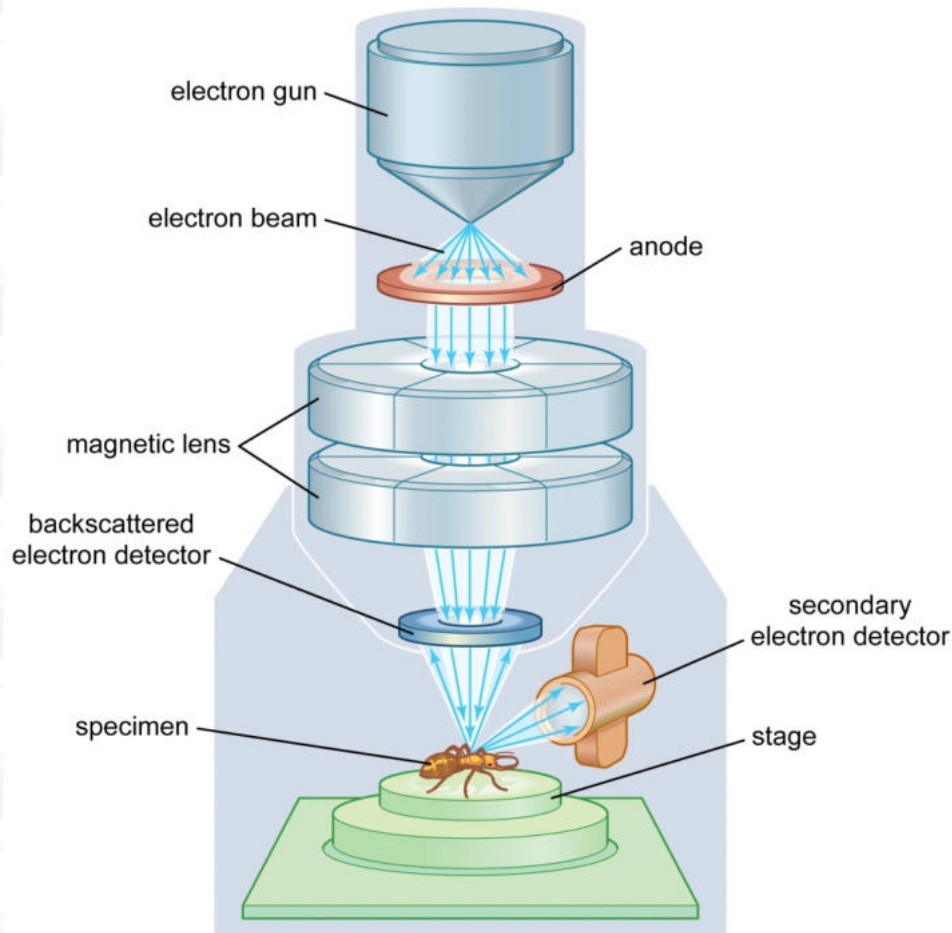
4. Energy-Dispersive X-ray Spectroscopy (EDS): With EDS detectors, electron viruses attacking a cell of bacterium microscopically analyze elemental composition (≤ 50000) by TEM by detecting X-rays emitted when high-energy electrons interact with the sample.

5. Materials Science: Crucial for studying the microstructure of materials, aiding understanding of relationships between microstructure and properties in metals, ceramics, and polymers.

6. Nanotechnology: Essential for imaging and characterizing nano-materials, contributing to the development of new materials and devices through observation and manipulation of viruses attacking a cell of bacterium nanoparticles and nanostructures (< 35000) by SEM.

7. Advancements in Medicine: Contributes to medical research by providing insights into the structure of viruses, bacteria, and cellular organelles, aiding understanding of diseases and therapeutic development.

8. Quality Control in Industry: Used across industries for quality control and failure analysis, helping identify defects, analyze material composition, and ensure product integrity at the microscopic level.



Uncertainty Principle:

The uncertainty principle mainly applied to experiments in physics labs, however some real-world effects are given.

Heisenberg Uncertainty Principle:

It states that it is impossible to measure simultaneously, certain pairs of properties of a subatomic particle, such as its position and momentum, with absolute certainty.

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

